

SOUVIK NATH

Python Full Stack Developer | Django | FastAPI | React.js | Next.js | TypeScript

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PROFESSIONAL SUMMARY

Results-driven Python Full Stack Developer with 2.8+ years of experience designing, building, and delivering enterprise-grade web applications across fintech and core banking domains. Proficient in Python, Django, Django REST Framework, FastAPI, microservices architecture, and modern frontend development with Next.js, React.js, and TypeScript. Experienced in SDLC best practices, Agile/Scrum methodologies, clean architecture, and scalable system design. Skilled in developing production-ready REST APIs, asynchronous processing with Celery and Redis, event-driven systems using Kafka, database optimization, and cloud deployment on Google Cloud Platform (GCP), with working exposure to AWS. Delivered two production-grade platforms—MvLEND, a composable core banking and lending platform, and Aura Finance, an AI-powered personal finance assistant.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL

Backend: Django, Django REST Framework (DRF), FastAPI, REST APIs, Microservices Architecture, Celery, Redis, WebSockets, Async Programming, Apache Kafka (event-driven)

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, shadcn/ui, React Query, Zustand

Databases: PostgreSQL, CockroachDB, MySQL, with a focus on schema design, query optimization, and performance tuning

DevOps & Cloud: Docker, Docker Compose, GitHub Actions, CI/CD Pipelines, Linux, Git & GitHub, Google Cloud Platform (GCP) as primary cloud environment, with working knowledge of AWS (EC2, S3)

Engineering: System Design, SDLC, Agile/Scrum, Distributed Systems, Caching, Performance Optimization, Debugging & Root Cause Analysis, Production Support, Code Review

Security: JWT Authentication, Role-Based Access Control (RBAC), OAuth2, Session Management, API Integration

Testing & Tools: Postman, Pytest

AI Integration: OpenAI API, AI Workflows, LLM, RAG, Prompt Engineering, Embeddings, Vector Search

PROFESSIONAL EXPERIENCE

Software Engineer, Python Full Stack | Gowdanar Technosoft Pvt Ltd, Bangalore Aug 2023 - Present

- Designed scalable full-stack web applications using Next.js, Django REST Framework, and FastAPI, supporting secure, high-performance fintech platforms for enterprise clients.
- Contributed to 40+ production-grade RESTful and asynchronous APIs for authentication, onboarding, and transaction processing, helping standardize API contracts used by frontend and third-party services.
- Optimized PostgreSQL queries, indexing strategies, and schema design for high-volume financial data, helping improve API response times.
- Engineered background and asynchronous processing pipelines using Celery, Redis, and WebSockets to support real-time updates across fintech applications.
- Applied clean architecture and modular microservices patterns across backend services, improving service isolation and maintainability.
- Implemented JWT authentication and RBAC-based access control for enterprise applications, aligned with SOLID design principles and internal security standards.
- Containerized backend services with Docker and Docker Compose, and supported CI/CD deployment pipelines via GitHub Actions, deploying primarily on Google Cloud Platform (GCP) with working exposure to AWS EC2 and S3.
- Automated third-party API integrations and internal backend processes, reducing manual effort and improving system reliability.

- Collaborated within Agile/Scrum teams through sprint planning, stand-ups, and retrospectives to deliver production-grade features on schedule.
- Handled debugging, root-cause analysis, and production support for critical fintech systems, partnering with product, QA, and DevOps teams to maintain stability.

KEY PROJECTS

MvLEND - Composable Core Banking & Lending Platform

Tech Stack: Django, DRF, FastAPI, Kafka, CockroachDB, Next.js, TypeScript, GCP

- Contributed to a digital lending platform supporting borrower onboarding, loan processing, repayment tracking, and fund disbursement.
- Worked on Kafka-based event-driven messaging to decouple services and support asynchronous transaction processing.
- Built REST APIs using Django REST Framework and FastAPI covering the loan lifecycle, including origination, repayment, and account management, secured with JWT authentication and RBAC.
- Developed responsive operational dashboards using Next.js and Tailwind CSS for monitoring, reporting, and account management.
- Helped design CockroachDB schemas for accounts and transactions, with a focus on query performance.

Aura Finance – AI-Powered Personal Finance Assistant

Tech Stack: Next.js, TypeScript, Django REST Framework, PostgreSQL, OpenAI API, Celery, Redis, pgvector, JWT, Cloudflare R2, Docker

Live Demo: ai-finance-assistant-phi.vercel.app | **GitHub:** github.com/souviknath18/ai-finance-assistant

- Developed a full-stack AI-powered personal finance platform using Next.js and Django REST Framework, enabling secure authentication, financial document processing, and intelligent transaction management.
- Built an AI-driven transaction extraction pipeline supporting PDF, CSV, and image-based financial documents using OpenAI, asynchronous processing with Celery and Redis, and hybrid rule-based parsing for improved accuracy.
- Implemented automatic transaction categorization, semantic transaction search, and similarity matching using OpenAI embeddings, pgvector, and Retrieval-Augmented Generation (RAG) techniques to provide personalized financial insights.
- Designed scalable RESTful APIs for uploads, transactions, budgeting, subscriptions, reports, and AI-powered search with JWT authentication, PostgreSQL, optimized queries, and Cloudflare R2 object storage.
- Developed a responsive dashboard with real-time upload tracking, AI processing status, financial analytics, and intelligent spending recommendations while deploying the application using Vercel and Railway.

EDUCATION

Bachelor of Technology in Electronics & Communication Engineering

Assam University, Silchar | Graduated: 2021